

Question Paper

Exam Date & Time: 22-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Blockchain Architecture Design and Use Cases [NPT_4128]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Evaluate the modular architecture of Hyperledger Fabric 2.x, highlighting how endorsement, validation, and commitment phases ensure transaction consistency. (3)
A) Course Outcome: "CO5" / Cognitive Level: "Evaluate"
B) Identify the function of Certificate Authority (CA) in Fabric and explain how it integrates with the MSP for decentralized identity verification. (3)
Course Outcome: "CO1" / Cognitive Level: "Application"
C) Distinguish Fabric's ordering mechanisms (Solo, Kafka, Raft) in terms of fault tolerance, scalability, and use cases. (4)
Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Examine the steps of defining assets, participants, and transactions in Hyperledger Composer. (3)
A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
B) Defend the process of connecting a Composer Business Network to Fabric through REST API integration. (3)
Course Outcome: "CO2" / Cognitive Level: "Evaluate"
C) Design a blockchain-based patient-record management system using Composer and discuss its security features. (4)
Course Outcome: "CO2" / Cognitive Level: "Create"
- 3) Analyze the transformation in cross-border remittance systems enabled by blockchain. (3)
A) Course Outcome: "CO4" / Cognitive Level: "Analysis"
B) Relate the role of stablecoins and tokenization in enhancing financial transparency. (3)
Course Outcome: "CO5" / Cognitive Level: "Application"
C) Inspect the impact of blockchain on compliance and anti-money laundering (AML) practices. (4)
Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4) Defend blockchain provenance and explain its role in supply-chain traceability. (3)
- A) Course Outcome: "CO4" / Cognitive Level: "Evaluate"
- B) Identify how IoT integration improves data authenticity in blockchain-based logistics networks. (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- C) Analyze IBM Food Trust as a case study of blockchain for food safety and consumer trust. (4)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 5) With a neat diagram, Appraise the end-to-end flow of deploying and invoking a smart contract in Ethereum, including gas usage and state updates. (4)
- A) Course Outcome: "CO3" / Cognitive Level: "Evaluate"
- B) Utilize Proof-of-Stake mechanism and contrast it with Proof-of-Work in terms of efficiency and security (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- C) Examine a Merkle Tree. Explain its role in blockchain transaction verification with an example. (3)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"

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Event Name		Subject	
058-MIT BLR-MAKEUP EXAM-2025-2026		NPT_4128:Blockchain Architecture Design and Use Cases	
#	Course Outcome	Description	Marks
1	1	CO 1	7
2	2	CO 2	10
3	3	CO 3	21
4	4	CO 4	6
5	5	CO 5	6
Total			50

* **NA** - CO Not Selected

Event Name		Subject	
058-MIT BLR-MAKEUP EXAM-2025-2026		NPT_4128:Blockchain Architecture Design and Use Cases	
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	4	12(24%)
4	Analysis	6	21(42%)
5	Create	4	13(26%)
6	Evaluate	1	4(8%)
Total		15	50 (100%)